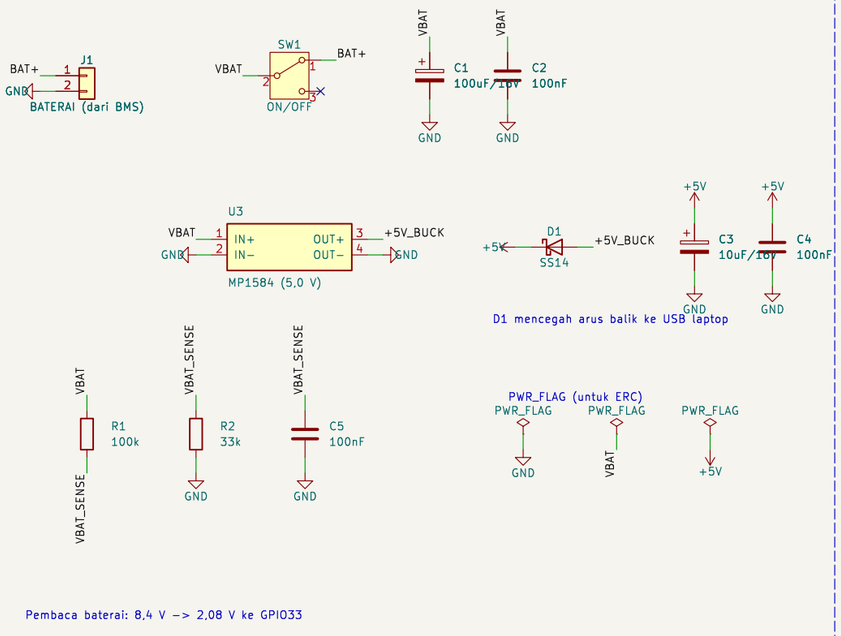
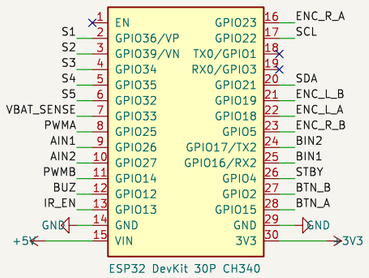
1. DAYA: baterai, saklar, buck 5 V, pembaca baterai

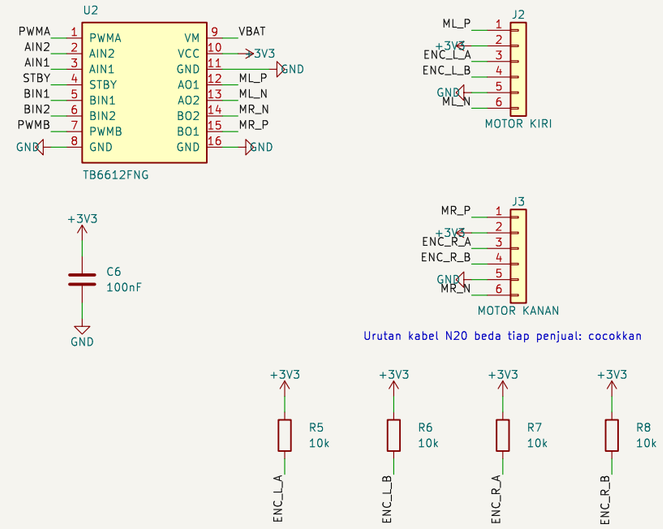


2. ESP32 DevKit 30 pin

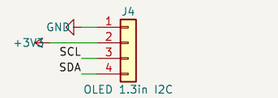


GPIO12 (BUZ) harus LOW saat boot: ada R4 10k di basis Q1

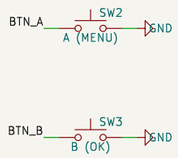
3. DRIVER MOTOR + KONEKTOR MOTOR



4. OLED + TOMBOL

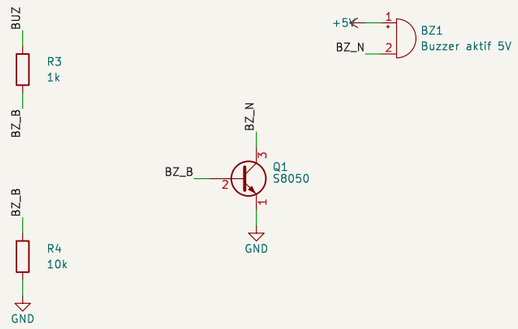


Sebagian OLED urutannya VCC–GND: cek



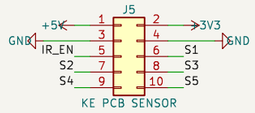
Pull-up internal ESP32 (INPUT_PULLUP)

5. BUZZER



Buzzer magnetik: tambah 1N4148 paralel

6. KE PCB SENSOR (J5 = lubang tembus)



Urutan pin sama persis dengan J1 PCB sensor

J5 tidak dipasang komponen: pin header J1 PCB sensor masuk dari bawah lewat lubang J5, disolder di atas

7. LUBANG BAUT M3



H1–H2 rangka, H3–H4 bahu, H5–H6 spacer PCB sensor

Format KiCad 7, dibuka KiCad 8/9 otomatis di-upgrade
Pin modul = susunan umum, CEK dengan modul yang dibeli
Acuan: docs/skematik-v1.md
P01 Kel.5 PjBL DTEO FV ITS

Sheet: /
File: ltm-main.kicad_sch

Title: LTM – PCB Utama

Size: A3	Date: 2026-10-11	**Rev: v1-starter**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1